

Multiple Choice

1. **Answer:** D

Options: A) Pyloric stenosis; B) Schizophrenia; C) Spina bifida (neural tube defects); D) Marfan syndrome

Note: Marfan syndrome is primarily caused by a single gene mutation in the FBN 1 gene, leading to a clear Mendelian inheritance pattern rather than multifactorial inheritance, which involves multiple genes and environmental factors.

2. **Answer:** D

Options: A) High density lipoprotein; B) HMG-CoA reductase; C) Low density lipoprotein; D) Low density lipoprotein receptor

Note: Familial hypercholesterolaemia is caused by mutations in the gene that encodes the low density lipoprotein receptor, which is crucial for the uptake of cholesterol from the bloodstream, leading to elevated cholesterol levels in affected individuals.

3. **Answer:** B

Options: A) A norovirus vaccine; B) Attention to handwashing and hygiene; C) Quarantine; D) New antiviral drugs

Note: Attention to handwashing and hygiene is crucial for controlling disease caused by norovirus because the virus is highly contagious and primarily spreads through contaminated hands, surfaces, and food.

4. **Answer:** A

Options: A) 46,XX,der(14;21)(q 10;q 10)pat+21; B) 47,XY,+13; C) 45,XX,rob,(14;21)(q 10;q 10); D) 46,XY,t(2;3)(q 21;q 12)

Note: The karyotype 46,XX,der(14;21)(q 10;q 10)pat+21 indicates the presence of an extra chromosome 21 due to a Robertsonian translocation, which is a common genetic cause of Down syndrome.

5. **Answer:** D

Options: A) HIV; B) Rubella teratogenicity; C) Microcephaly and Zika; D) Hepatitis C

Note: Hepatitis C was identified through molecular techniques and blood screening assays rather than through initial clinical observation, distinguishing it from other viral infections that were first recognized based on clinical symptoms.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The aorta is a major artery that carries oxygenated blood from the heart to the rest of the body, not air to the lungs.

7. **Answer:** True

Options: A) True; B) False

Note: Facing older adults with hearing loss allows them to utilize lip-reading as a complementary communication strategy, enhancing their understanding of spoken language.

8. **Answer:** True

Options: A) True; B) False

Note: A homeotic mutation occurs when a gene that regulates the identity of body parts is altered, leading to the transformation of one anatomical structure into another during the organism's development.

9. **Answer:** True

Options: A) True; B) False

Note: Adenoviruses do not integrate their genetic material into the host genome, which significantly reduces the risk of insertional mutagenesis and enhances their safety for long-term gene expression in gene therapy.

10. **Answer:** True

Options: A) True; B) False

Note: Recombinant alpha-iduronidase is an enzyme replacement therapy that compensates for the deficiency of the enzyme in individuals with Hurler syndrome, which helps to alleviate the symptoms of this lysosomal storage disorder.

Long Form Response

11. **Answer:** Young survivors of significant loss, such as the death of a parent or close family member, face a higher likelihood of developing depression due to several interrelated factors. These include the profound emotional impact of grief, the disruption of family dynamics, and the potential for feelings of isolation among peers who may not understand their experience. Additionally, developmental stages in adolescence and young adulthood can exacerbate vulnerability, as these individuals are still forming their identities and coping mechanisms. The implications for mental health support are significant; it is crucial to provide targeted interventions that address both the emotional and psychological needs of these survivors, fostering resilience and promoting healthy coping strategies. Early and compassionate support can mitigate the risk of lasting mental health issues.

Note: correct because it identifies the emotional, social, and developmental factors that increase the risk of depression in young survivors of significant loss, while also recognizing the need for tailored mental health support to address these unique challenges.

12. **Answer:** Clonal expansion is a crucial process in the adaptive immune system, where specific lymphocytes, such as B cells and T cells, proliferate in response to an antigen. This expansion enables a rapid and robust immune response, as pre-existing antibodies and T cells that recognize the pathogen multiply exponentially, increasing their numbers significantly. The effectiveness of this response lies in the ability of these expanded clones to target and eliminate pathogens efficiently, providing both immediate protection and the formation of memory cells for long-term immunity. Consequently, clonal expansion not only enhances the adaptive immune system's capacity to respond to infections but also plays a vital role in the development of immunological memory, ensuring quicker and more effective responses to subsequent exposures to the same pathogen.

Note: correct because it accurately describes clonal expansion as a vital mechanism that enhances the adaptive immune response by allowing specific lymphocytes to proliferate and effectively target pathogens, thereby increasing the overall efficacy of the immune system.